

1.8

1_1010_1111 and 1AF 、 b is faster

1.9

(a)* $(10110.0101)_2$

(c)* $(26.24)_8$

(e) $(1010.1101)_2$

(b)* $(16.5)_{16}$

(d) $(DADA.B)_{16}$

a-c 22.3125

Step 4

4 of 6

(d)

$$\begin{aligned}(DADA.B)_{16} &= 13 \times 16^3 + 10 \times 16^2 + 13 \times 16^1 + 10 \times 16^0 + 11 \times 16^{-1} \\ &= \boxed{(56026.6875)_{10}}\end{aligned}$$

Step 5

5 of 6

(e)

$$\begin{aligned}(1010.1101)_2 &= 1 \times 2^3 + 1 \times 2^1 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-4} \\ &= \boxed{(10.8125)_{10}}\end{aligned}$$

1.10

1.10 (a) $1.10010_2 = 0001.1001_2 = 1.9_{16} = 1 + 9/16 = 1.563_{10}$

(b) $110.010_2 = 0110.0100_2 = 6.4_{16} = 6 + 4/16 = 6.25_{10}$

Reason: 110.010_2 is the same as 1.10010_2 shifted to the left by two places.

1.13

A 11011.0101_2

B. $.666666667_{10} \approx .10101010_2 = .5 + .125 + .0313 + .0078 = .6641_{10}$

$.10101010_2 = .1010_2 .1010_2 = .AA_{16} = 10/16 + 10/256 = .6641_{10}$ (Same as (b)).

the result $.6640625_{10}$ is close but not exact

C. $AA_{16} = 10/16 + 10/16 = .625 + .0390625 = .6640625_{10}$

the answer is the same as in (b)

1.14 求下列二进制数的反码和补码：

(a) 0001_0000 1110_1111 1111_0000

(b) 0000_0000 11111111 00000000

(c) 1101_1010 00100101 00100110

(d) 1010_1010 01010101 01010110

(e) 1000_0101 01111010 01111011

(f) 1111_1111 00000000 00000001

1.17

(a) $4,637 - 2,579$

(b) $125 - 1,800$

(c) $2,043 - 4,361$

(d) $1,631 - 745$

A $4637 - 2579 = 2058$ B $125 - 1800 = -1675$

C $2043 - 4361 = -2318$ D $1631 - 745 = 886$

1.35

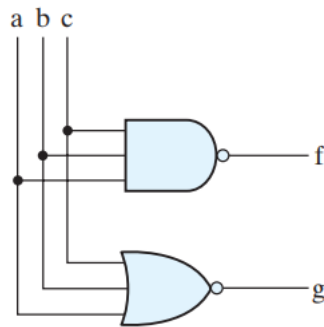
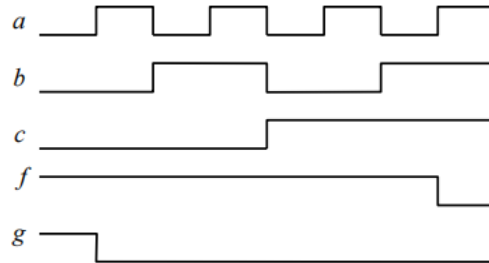


FIGURE P1.35



1.36

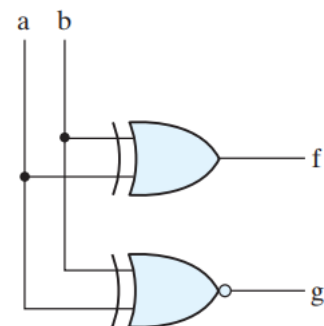
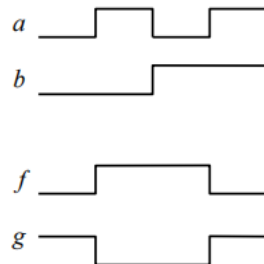


FIGURE P1.36



2.2ef

(a)* $xy + xy'$

(c)* $xyz + x'y + xyz'$

(e) $(a + b + c')(a'b' + c)$

(b)* $(x + y)(x + y')$

(d)* $(A + B)'(A' + B)'$

(f) $a'bc + abc' + abc + a'bc'$

(a) $xy + xy' = x(y + y') = x$

(b) $(x + y)(x + y') = x + yy' = x(x + y') + y(x + y') = xx + xy' + xy + yy' = x$

(c) $xyz + x'y + xyz' = xy(z + z') + x'y = xy + x'y = y$

(d) $(A + B)'(A' + B) = (A'B')(A + B) = (A'B')(AB) = A'(B'B)A = 0$

(e) $(a + b + c')(a'b' + c) = aa'b' + ac + ba'b' + bc + c'a'b' + c'c = ac + bc + a'b'c'$

(F) B

2.4e

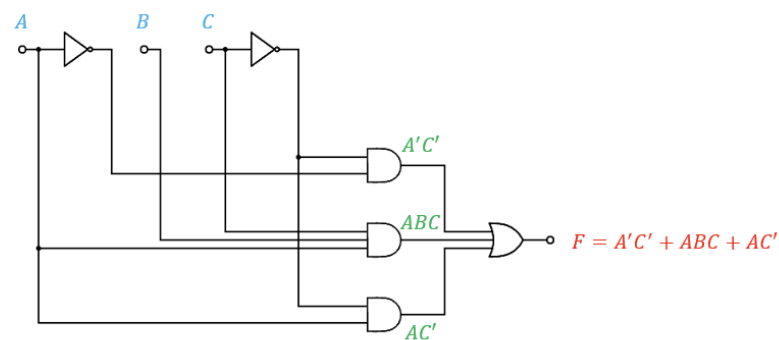
- (a) * $A'C' + ABC + AC'$
 (b) * $(x'y' + z)' + z + xy + wz$
 (c) * $A'B(D' + C'D) + B(A + A'CD)$
 (d) * $(A' + C)(A' + C')(A + B + C'D)$
 (e) $ABC'D + A'BD + ABCD$

to three literals
to three literals
to one literal
to four literals
to two literals

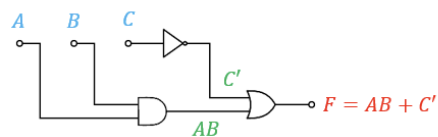
$$ABCD + A'BD + ABC'D = ABD + A'BD = BD$$

2.7

AA

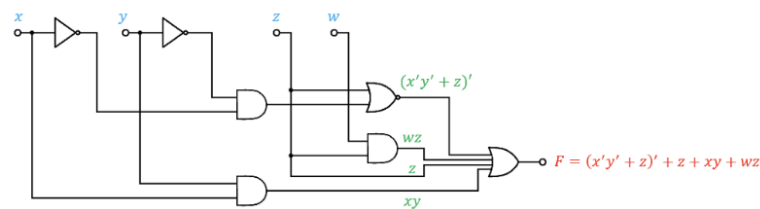


Original

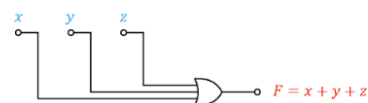


Simplified

BB



Original



Simplified

CC

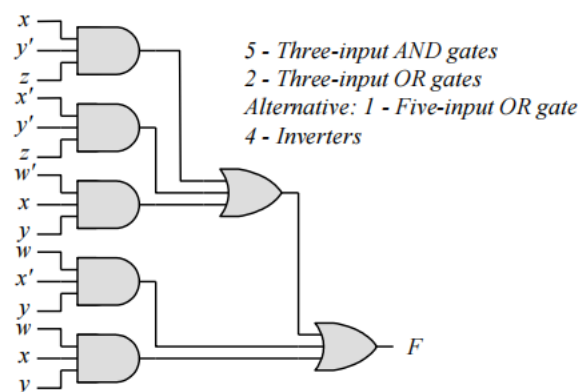
2.18

$$F = xy'z + x'y'z + w'xy + wx'y + wxy$$

2.18 (a)

wx y z	F	$F = xy'z + x'y'z + w'xy + wx'y + wxy$ $F = \Sigma(1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$
00 0 0	0	
00 0 1	1	
00 1 0	0	
00 1 1	0	
01 0 0	0	
01 0 1	1	
01 1 0	1	
01 1 1	1	
10 0 0	0	
10 0 1	1	
10 1 0	1	
10 1 1	1	
11 0 0	0	
11 0 1	1	
11 1 0	1	
11 1 1	1	

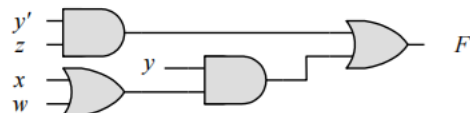
(b)



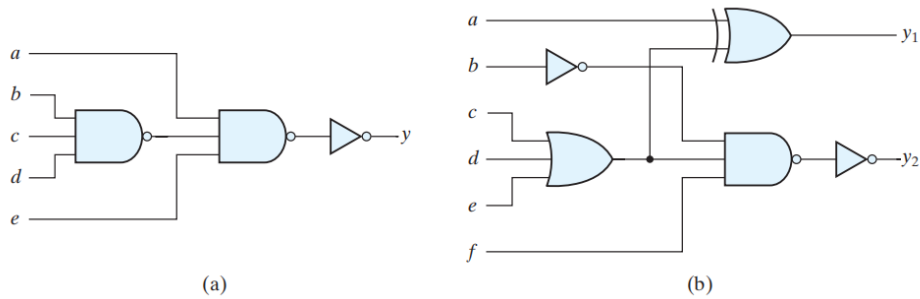
(c) $F = xy'z + x'y'z + w'xy + wx'y + wxy = y'z + xy + wy = y'z + y(w + x)$

(d) $F = y'z + yw + yx = \Sigma(1, 5, 9, 13, 10, 11, 13, 15, 6, 7, 14, 15)$
 $= \Sigma(1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$

(e)



2.28



2.28 (a) $y = a(bcd)'e = a(b' + c' + d')e$

$$y = a(b' + c' + d')e = ab'e + ac'e + ad'e \\ = \Sigma(17, 19, 21, 23, 25, 27, 29)$$

a bcde	y	a bcde	y
0 0000	0	1 0000	0
0 0001	0	1 0001	1
0 0010	0	1 0010	0
0 0011	0	1 0011	1
0 0100	0	1 0100	0
0 0101	0	1 0101	1
0 0110	0	1 0110	0
0 0111	0	1 0111	1
0 1000	0	1 1000	0
0 1001	0	1 1001	1
0 1010	0	1 1010	0
0 1011	0	1 1011	1
0 1100	0	1 1100	0
0 1101	0	1 1101	1
0 1110	0	1 1110	0
0 1111	0	1 1111	0

(b) $y_1 = a \oplus (c + d + e) = a'(c + d + e) + a(c'd'e') = a'c + a'd + a'e + ac'd'e'$

$$y_2 = b'(c + d + e)f = b'cf + b'df + b'ef$$

$$y_1 = a(c + d + e) = a'(c + d + e) + a(c'd'e') = a'c + a'd + a'e + ac'd'e'$$

$$y_2 = b'(c + d + e)f = b'cf + b'df + b'ef$$

$a'-c---$	$a'--d--$	$a'---e-$	$a-c'd'e'-$
001000 = 8	000100 = 8	000010 = 2	100000 = 32
001001 = 9	000101 = 9	000011 = 3	100001 = 33
001010 = 10	000110 = 10	000110 = 6	110000 = 34
001011 = 11	000111 = 11	000111 = 7	110001 = 35

001100 = 12	001100 = 12	001010 = 10
001101 = 13	001101 = 13	001011 = 11
001110 = 14	001110 = 14	001110 = 14
001111 = 15	001111 = 15	001111 = 15

$-b'c--f$	$-b'-d-f$	$-b'--ef$
011000 = 24	010100 = 20	010010 = 18
011001 = 25	010101 = 21	010011 = 19
011010 = 26	010110 = 22	010110 = 22
011011 = 27	010111 = 23	010111 = 23
011100 = 28	011100 = 28	011010 = 26
011101 = 29	011101 = 29	011001 = 27
011110 = 30	011110 = 30	011110 = 30
011111 = 31	011111 = 31	011111 = 31
101001 = 9	101001 = 9	101001 = 41
101011 = 11	101011 = 11	101011 = 43
101101 = 13	101101 = 13	101101 = 45
101111 = 15	101111 = 15	101111 = 47
100011 = 7	100011 = 7	100011 = 35
100111 = 11	100111 = 11	100111 = 39
100111 = 15	100111 = 15	100111 = 51
100011 = 35	100011 = 35	100011 = 55